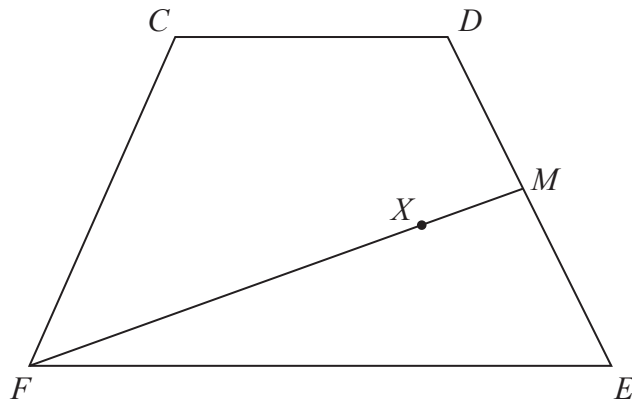


20 $CDEF$ is a quadrilateral.



$$\overrightarrow{CD} = \mathbf{a}, \overrightarrow{DE} = \mathbf{b} \text{ and } \overrightarrow{FC} = \mathbf{a} - \mathbf{b}.$$

- (a) Express $\overrightarrow{FE}$ in terms of $\mathbf{a}$ and/or $\mathbf{b}$.
Give your answer in its simplest form.

(2)

M is the midpoint of DE .

X is the point on FM such that $FX:XM = n:1$

CXE is a straight line.

- (b) Work out the value of n .

$$n =$$

(4)

(Total for Question 20 is 6 marks)